Who Owns Academic Knowledge? The Evolution of Intellectual Property Rights in American Universities

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Definitions, assignment and creator protections in historical policy comparisons
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Research status: the frozen archive is reproduced. Source-linked comparisons at two universities identify changes in selected stated relationships, including mixed changes and contractual precedents. Selection is purposive, historical coverage is incomplete, and no independent human legal validation has been obtained. This manuscript is not designated submission-ready.

Abstract

When universities name new kinds of intellectual property, do they claim new rights? We distinguish definitions, assignment mechanisms and creator protections in a historical corpus of 480 observed policy vintages from 150 American universities, 1944–2025. The archive contains 330 adjacent within-university comparisons. Source-linked case comparisons show why their lexical changes cannot measure ownership expansion. Utah's 1999–2013 policies combine more explicit categories and present-assignment wording with a narrower resource trigger. Hawai'i's 1985 and 2020 endpoints preserve qualified creator ownership while specifying instructional uses; a faculty agreement already addressed distance-learning control in 2003–2009. These selected comparisons demonstrate changes in separate dimensions of control, including provisions pointing in different directions within an institution. They do not estimate national transition frequencies or enforceability. We provide reproducible corpus results, a relationship-level evidence ledger, and explicit distinctions between authenticated provisions, interpretive uncertainty and missing independent validation.

Keywords: university intellectual property; ownership; assignment; institutional change; academic knowledge; legal text.

1 Introduction

In 1999, the University of Utah asserted title to inventions arising from university employment, research or specified uses of its resources, and required inventors to execute assignments when requested. Its 2013 policy used express present-assignment language and separately defined covered research outputs. Yet the revision also expressly excluded ordinary departmental resources from its resource-use trigger. These provisions do not move together on a single scale of institutional control. A more explicit transfer mechanism can coexist with a narrower ground for claiming an invention (University of Utah, 1999, III.B.1) (University of Utah, 2013, II.B–C and III.B.1).

A different problem appears in Hawai'i's policies. The 2020 executive policy expressly discusses distance learning and regulates recording and later use, while retaining a qualified creator-ownership baseline for academic materials. But an earlier faculty agreement already addressed the same kinds of use. Reading the newer policy alone would mistake the appearance of terminology in one instrument for the beginning of institutional governance over the activity. Neither the persistence of creator title nor the addition of a use authorization can be understood by counting category names (University of Hawaii, 1985, 2020; University of Hawaii Professional Assembly and Board of Regents, 2003).

This article asks when an expansion in the named objects of intellectual property corresponds to a change in the rights universities claim. The relevant distinction is not simply broad versus narrow

ownership. A policy can alter the objects covered, the people bound, the circumstances triggering a claim, the means of assignment, or the uses reserved to creators. Those changes may reinforce one another, offset one another, or address separate problems. They also concern stated institutional arrangements, which need not establish enforceable title in a particular dispute.

The historical corpus comprises 480 observed institution-year policy records from 150 American universities. We reproduce its population and compare terminology across 330 adjacent observed vintages. We then examine selected operative relationships at two institutions for which official historical material could be recovered. Utah supplies one adjacent comparison. Hawai'i supplies a longer endpoint comparison spanning an intervening, incompletely authenticated 2014 record. We retain that distinction rather than add the longer comparison to the adjacent-pair denominator. The case selection is purposive; it supports evidence about mechanisms and counterexamples, not estimates of how often American universities expanded their claims.

Prior scholarship has examined ownership regimes and already distinguishes ownership, control and income rights (Kenney and Patton, 2011; Halilem et al., 2017). Jamali (2024) codes definitions, creator groups and allocation arrangements across Australian university policies. The missing observation addressed here is narrower: how different elements of a university's stated arrangement change between historical sources, and whether a newly named object is connected to a newly stated entitlement. Our contribution is a source-linked account of that relationship, nested in a reproducible American archive. It is not the discovery that legal definitions and ownership differ, nor a completed national census of ownership-rule changes.

The evidence establishes that an institutional revision need not have a uniform direction. Utah's more explicit claim and assignment provisions coexist with qualifications on resource-based control; Hawai'i's more detailed treatment of teaching materials preserves creator ownership and divides use rights. Earlier contractual provisions further complicate the dating of apparent changes. These findings differ from the proposition that policy vocabulary alone expanded, while leaving the population distribution of substantive transitions unresolved. They also distinguish this paper from studies of policy communication and commercialization: we neither reconstruct a communication index nor use patent or licensing outcomes as substitutes for measuring allocation.

2 Property rights, scientific work and institutional rules

Ownership matters in incomplete-contracting accounts because a contract cannot specify every future use of an asset. Residual authority affects the parties' incentives and their positions when unforeseen circumstances arise (Grossman and Hart, 1986; Hart and Moore, 1990; Hart, 1995). Applied to academic knowledge, this perspective suggests asking who may authorize a license, refuse a transfer, release an invention, or continue using a work after title is assigned. It does not

imply that the broadest university claim is efficient. The relevant investments include research effort, disclosure, development and access by other researchers; different allocations may favor different investments.

Knowledge also differs from the physical assets that motivate many property-rights models. Information can be used by multiple researchers, legal protection may attach to only part of a research output, and publication can coexist with restrictions on materials or commercial exploitation. The economics of open science emphasizes priority, disclosure and reputation as organizing institutions (Dasgupta and David, 1994). Academic freedom can have value precisely because researchers retain discretion over lines of inquiry that a commercial principal would not immediately pursue (Aghion et al., 2008). A policy that reserves educational use to a creator while assigning commercialization authority to the university therefore creates a divided arrangement, not an all-or-nothing transfer of control.

Institutional rules mediate these choices, but written rules are only one part of institutional change (North, 1990). Administrators may formalize an existing practice, respond to a funding condition, consolidate several policies, or alter the people to whom a rule applies. A longer definition could resolve uncertainty without changing an operative allocation. Conversely, deleting a student exception could alter the scope of a claim even if every category name remains unchanged. These possibilities motivate observable distinctions rather than a single formal prediction of expansion.

The historical literature cautions against treating 1980 as the origin of university patent governance. Palmer (1962) collected university patent policies well before Bayh–Dole. Studies of twentieth-century patenting locate the legislation within earlier administrative developments and changing research opportunities (Mowery et al., 2001; Sampat, 2006). Organizational accounts likewise trace institution-building and the normalization of technology-transfer practices before the statute (Popp Berman, 2008; Colyvas, 2007). A before-and-after difference at 1980 would confound those developments with legal change, archival availability and changes in research composition. We do not use Bayh–Dole as a natural experiment.

Normative arguments are contested. Kenney and Patton (2009) criticize the institutional invention-ownership model, while the National Research Council frames technology transfer as an institutional responsibility that should serve broader public purposes (National Research Council, 2011). Comparative European evidence shows that ownership reforms occur alongside other organizational changes (Geuna and Rossi, 2011); the Italian experience similarly links policy development and university patenting without establishing that the same mechanism applies to American institutions (Baldini et al., 2006). These sources support competing interpretations of institutional claims, not a presumption that more claims are either beneficial or harmful.

The incentive literature also gives reason to distinguish allocation from outcomes. Lach and

Schankerman (2008) report a relationship between royalty incentives and university invention, whereas Ouellette and Tutt (2020) identify coding problems and challenge the robustness of the earlier evidence. Even a reliable description of royalty shares would not establish who initially owns a work. Nor would an association between a policy and patents identify the effect of its assignment clause. Our choice to analyze rights directly avoids using commercialization as a substitute for legal measurement.

The scientific-commons literature identifies another set of competing mechanisms. Fragmented exclusion rights may obstruct subsequent research (Heller and Eisenberg, 1998), and policy scholarship questions whether privatization always advances publicly funded research (Eisenberg, 1996; Rai and Eisenberg, 2003; Nelson, 2004). Yet a patent can sometimes facilitate the search for a developer (Hellmann, 2007). Empirical studies distinguish citation effects, material-transfer barriers and changes in research access (Murray and Stern, 2007; Walsh et al., 2005; Murray et al., 2016). Those outcomes are not interchangeable. A license to use a material, title to a patent, and authority over a database can affect scientific activity through different channels. Scientific norms may further qualify how formal rights are exercised (Rai, 1999).

For data, the distinction is especially consequential. Research-data sharing involves heterogeneous objects, responsibilities and incentives (Borgman, 2012). Contractual arrangements can support a research commons even where proprietary claims are available (Reichman and Uhlir, 2003). FAIR principles address the management and reuse of data; they do not by themselves allocate ownership (Wilkinson et al., 2016). An increase in references to databases could therefore reflect stewardship, disclosure requirements, licensing, or ownership. Only the surrounding rule establishes which interpretation is supported.

3 What a policy can establish legally

A university policy is an institutional statement of a claimed rule. Its legal effect depends on the underlying right and on matters that a text corpus usually does not observe: applicable law, the employment or student relationship, assent, executed agreements, competing assignments, and the facts of creation. We separate an institutional ownership presumption from an assignment obligation and from wording that purports to make a present assignment. We also separate each of these from a judicial determination of title.

The distinction is visible in patent law. The Bayh–Dole framework addresses rights in inventions made with federal assistance, including a contractor's election to retain title to qualifying subject inventions (United States Congress, 2024). In *Stanford v. Roche*, the Supreme Court held that the Act does not automatically vest title in federal contractors or authorize them unilaterally to take an inventor's rights (Supreme Court of the United States, 2011, pp. 785–789). The Court

did not review the lower court's construction of the competing assignment language. We therefore do not treat the decision as establishing that every occurrence of a particular verb formula guarantees an effective transfer. Earlier employee-invention doctrine also distinguishes obligations arising from being hired to invent from other employment relationships and from an employer's shop rights (Supreme Court of the United States, 1933, pp. 187–189). Historical scholarship documents changes in how employee-inventor interests were understood (Fisk, 1998).

Copyright has a different structure. Initial ownership, work made for hire, and transfers are addressed in distinct statutory provisions. Ownership of a physical object is separate from ownership of the copyright embodied in it, and a transfer of copyright ownership generally requires a signed writing (United States Congress, 1976, sections 101, 201, 202 and 204). The agency inquiry used to determine employee status under work-made-for-hire doctrine cannot be replaced by a university's preferred label (Supreme Court of the United States, 1989). The position of academic and other creative employees raises additional questions about the relationship between employment and creative autonomy (Dreyfuss, 1987). We code what a policy reserves or claims, without resolving these external legal questions from its language alone.

A broad definition also cannot create an intellectual property right that applicable law does not recognize. Facts are not protected by copyright, although an original selection or arrangement may qualify (Supreme Court of the United States, 1991). A statement that an institution owns research data could concern records, access, contractual obligations, a protected compilation, or several of these at once. It does not establish an exclusive copyright in every underlying fact. Similarly, the Copyright Office's analysis of artificial intelligence emphasizes human authorship and the nature of the human contribution (United States Copyright Office, 2025). That report is agency analysis, not a new ownership statute. Mentioning AI in a policy cannot settle copyrightability or eliminate the need to identify the relevant object.

These distinctions generate a practical classification scheme. Each relationship has a rights-holder, a legal operation, a knowledge category, a trigger and a set of qualifications. The same provision may assign a university title while reserving a creator's noncommercial license. A broad resource-based presumption may coexist with an academic-work exception. Mixed and indeterminate cases are retained. The scheme is designed to test whether a change in definition reaches an operative rule, rather than assume that it does.

4 Archive construction and research design

The analysis begins with the original cleaned-sentence file from the Paper 1 replication repository. We pin its Git commit and SHA-256 checksum, together with the institution-year panel used only to distinguish direct observations from carried-forward records. The primary population remains

1944–2025. A single 1925 row is excluded; no other population repair is silently applied. The included file contains 87,159 cleaned sentences, 480 institution-year records and 518 institution-year-filename groups. The larger panel contains 4,277 rows, of which 3,797 are carried forward. These are not additional policy documents.

Table 1: Frozen analytical population

Quantity	Count
Universities	150
Observed institution-years	480
Institution-year-filename groups	518
Included cleaned sentences	87,159
Panel rows (not documents)	4,277
Carried-forward rows	3,797
Original-screen sentences	2,779
Records with screen matches	449
Adjacent observed pairs	330
Universities with pairs	129

Notes: The primary population is unchanged from the established 1944–2025 sample. A filename group is an archival unit, not a verified unique policy or revision. Panel identifiers are used only for population reconciliation.

The audit assigns each sentence an identifier based on its immutable source row, rather than rely on sentence numbers being unique. It retains raw-text and normalized-text hashes and records institution, vintage label, filename and sentence number. Normalization is limited to case and whitespace for duplicate diagnostics. It does not modify the primary analysis population. Reconstructed context files preserve the available cleaned sentences, including collisions, but are explicitly labeled as reconstructions: missing headings, original paragraphs and external cross-references cannot be recovered merely by joining them.

Documentary structure is consequential. There are 29 institution-years with more than one filename. A university-year can combine a substantive policy with forms or contain documents for different campuses. The Wake Forest 2017 record, for example, includes both medical-center and Reynolda-campus files. Treating the pooled record as one unified legal instrument would obscure the population to which each rule applies. The audit also finds 3,662 rows involved in repeated sentence-number keys, concentrated in Georgia Tech, and 2,703 repeated normalized sentences within filename groups. Fifteen document groups participate in exact normalized-text duplication. Such repetition is retained in the primary corpus and examined in sensitivity analysis, rather than interpreted as independent legal evidence.

Historical coverage is highly uneven (Figure 1). The earliest decades contain very few observed institution-years and no observations occur in the 1950s. Absence from the archive does not estab-

lish absence of a policy. Nor does a vintage label prove when a provision became effective. One filename contains a year inconsistent with its upstream record, and a pair of Delaware vintages has the same normalized sentence set. The database therefore supports comparisons between observed textual endpoints, not a continuous census of policy revisions.

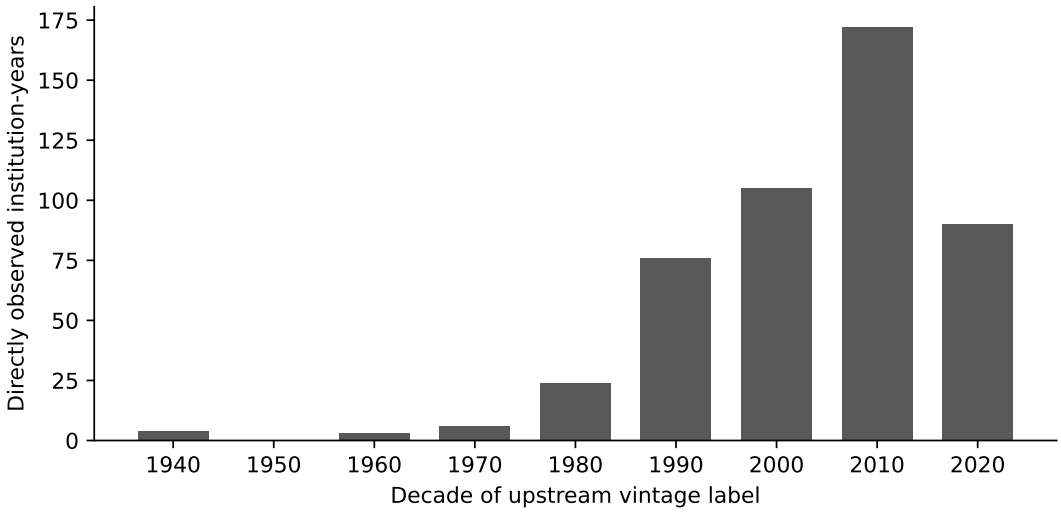


Figure 1: Observed institution-years by decade of upstream vintage label

We first reproduce the original sentence screen without changing its expressions. A sentence is a candidate when it contains both a named knowledge category and one of the specified rule patterns. These include exceptions and resource conditions as well as ownership and assignment language. A candidate is consequently not a positive classification of university ownership. The screen yields 2,779 sentences in 449 records. The remaining records cannot be classified as lacking ownership rules: they may use terms omitted by the screen or distribute an operative rule across sentences.

A second set of transparent expressions provides a broader review queue. It searches operations without requiring an asset term in the same sentence and records separate flags for nine knowledge categories, assignment wording, licenses, exceptions, procedures and triggers. Every output is marked as a machine-generated review aid and legally unadjudicated. We do not infer a rights-holder from the presence of an institution's name. The broader queue improves retrieval coverage for some inspected examples but has not been shown to improve precision or recall against a reference standard.

For historical description, we compare adjacent observed vintages within each university. This yields 330 pairs from 129 universities. For each category, the measure is whether any included sentence contains its specified terminology. The transition matrix records absence or presence at each endpoint. It is a matrix of mentions, not ownership. A separate estimate averages the adjacent-

pair change within each university and then gives each university equal weight. Percentile intervals resample those university means using 5,000 draws and a fixed seed. These intervals describe variation within the observed archive; they do not account for nonrandom archival selection, legal misclassification or date uncertainty.

No annual interpolation enters these comparisons. The intervals between observed vintages vary, and their endpoints do not locate the timing of an intervening change. Thirty-five pairs change the number of constituent files. Even where file counts are unchanged, document authority or campus scope may differ. The within-university design reduces compositional differences across institutions but cannot solve these documentary problems. For that reason, we do not estimate a time trend in legal expansion or attach causal meaning to the lexical transitions.

The substantive exercise uses a different unit: a relationship between an actor, an object, a legal operation and a condition. Thirteen selected relationships are compared across two institutional endpoint pairs. Each entry records both source versions, sections, immutable corpus row identifiers, short corpus excerpts, source hashes, interpretation and unresolved qualifications. The coding is disclosed AI-assisted close reading. It is not a trained classifier, independent human adjudication or an exhaustive inventory of either university's legal arrangements.

We authenticate sources at the level needed for the stated comparison. Utah's official archive supplies the March 1999 policy under its former number, 6-4. Its current revision history identifies the July 2013 successor, 7-002. The downloaded successor incorporates later editorial updates, so we compare selected operative passages against both the frozen 2013 corpus and the contemporaneous committee memorandum. We do not represent the downloaded PDF as pristine 2013 bytes. For Hawai'i, the official March 1985 executive policy and the December 2020 amended policy match the selected endpoint provisions. The latter retains an October 2014 effective-date header. Its intervening corpus record has not been independently authenticated. We therefore describe an endpoint difference, not a single revision enacted in 2020.

Comparison requires a common actor and fact pattern. A narrower resource trigger matters for a privately developed invention only if another trigger does not independently support institutional ownership. An exception for nonemployee students cannot be compared with an employee rule as though they governed identical people. We distinguish demonstrable wording changes from changes whose incremental legal scope remains uncertain. The register preserves all 330 adjacent comparisons: one has selected source-linked relationships examined here, and 329 remain contextually unclassified. The additional Hawai'i comparison is separately identified. No unreviewed pair is coded unchanged.

5 Results: changes in allocation rather than vocabulary alone

The clearest substantive result is the coexistence of changes pointing in different directions within the same institution. Utah's 1999–2013 comparison changes the stated assignment mechanism, resource conditions, student provisions and release timing. Hawai'i's longer comparison preserves a qualified creator baseline while adding particular instructional uses and changing other conditions. Table 2 presents selected comparisons; the complete ledger retains thirteen relationships and the source qualifications for each. Its rows are legal dimensions, not thirteen independent revisions.

Table 2: Selected changes in stated relationships

Relationship	Earlier endpoint	Later endpoint
Utah: Assignment mechanism	Claim of title plus assignment on request.	Express present-assignment wording plus confirmatory assignment on request.
Utah: Resource threshold	Resource trigger excludes facilities and materials available to the public.	Non-incidental trigger also excludes ordinary departmental resources; paid inventor time counts.
Utah: Release after inactivity	Creator claim honored after three years without active promotion or development.	Creator claim honored after a reasonable time without active promotion or development.
Hawai'i: Academic-work ownership	Qualified faculty/staff copyright baseline and own-initiative educational materials belong to creators.	Qualified employee copyright baseline and own-initiative educational materials remain creator-owned.
Hawai'i: Syllabi and reuse	General own-initiative educational-materials protection.	Syllabi classified as own-initiative materials; required accreditation copies confer no publication/reuse entitlement without consent.
Hawai'i: Distance-learning control	General educational-materials and commissioned-work provisions; faculty recording arrangement already appears in 2003-2009 CBA.	Compensated/release-time distance instruction may be recorded; later sale or retransmission requires written agreement.
Hawai'i: Sponsored publication delay	Patent-filing delay ceiling of six months from notice.	Patent-filing delay ceiling of two months from notice.

Notes: Purposive comparisons, not population frequencies. Utah is an adjacent 1999–2013 pair; Hawai'i compares 1985 and 2020 endpoints across an intervening unverified record. AI-assisted interpretations require independent review. See the evidence ledger for conditions, source sections and ambiguity; shortened table entries do not replace those qualifications.

Utah's earlier title assertion was already accompanied by an obligation to assign on request. The later wording adds a purported present assignment and retains the obligation to confirm it. We code a change in transfer mechanism, not a transition from inventor ownership to university ownership. Nor do we infer that a policy statement alone effected a legally valid transfer. This comparison

would disappear in a binary indicator recording whether either policy asserted university ownership. It also need not be accompanied by a new asset name: the change concerns how the policy claims rights, not only what it names.

The resource comparison points in another direction. In 1999, the resource ground excluded facilities, equipment or materials available to the public. The later definition also excludes resources ordinarily made available to an inventor and colleagues in the same unit, subject to its stated conditions, while counting compensated inventor time. Consider a privately developed invention outside an employee's university expertise and research, produced on unpaid time using only ordinary departmental equipment that is not available to the public. Holding those facts fixed, the revised resource branch is narrower. This is a conditional comparison: employment-related expertise, university research or other qualifying resources can independently preserve a claim. It would be wrong to report a general retreat from institutional ownership on this evidence.

Other Utah changes resist a simple direction. The earlier employment-or-research formulation becomes a more explicit combination of expertise, university research and non-incidental resources. Some activity within expertise may already have resulted from employment; the earlier wording does not justify coding every such activity as previously creator-owned. The student exception is recast around university research and incidental resources, with mere advice to independent student research distinguished from faculty performance or management of research. The earlier general resource claim and student exception also require interpretation together. We retain a mixed or indeterminate code rather than infer a uniform loss of student rights.

Release provisions provide a useful check against reading legislative summaries in isolation. The 1999 policy already promised to honor an inventor's claim after three years without active promotion or development. The later policy uses a reasonable-time standard. The change is real, but it is neither the first appearance of a release provision nor unambiguously a stronger creator right. A reasonable period may be shorter or longer than three years. Likewise, the earlier mixed-activity presumption covers faculty, staff and students, whereas the corresponding later paragraph expressly addresses students. That narrows the stated population of the presumption without eliminating employee ownership grounds elsewhere in the policy.

Utah's definition does explicitly enumerate data, biological materials and other research outputs, and brings software within the invention-policy framework. But software had already appeared in the 1999 commercialization provision. New placement inside an operative definition differs from the first appearance of the word, and neither establishes that every newly listed output was previously outside university claims. The notebook provision also combines institutional delivery requirements with faculty custody and a retained copy. The relationship is partly about possession and continued access; it cannot be reduced to an exclusive property right in underlying facts.

The contemporaneous memorandum is informative about institutional motivation. Prestwich

and Morris (2013) describe a review prompted partly by *Stanford v. Roche*, alongside disputes about scope, changes in university operations and other administrative concerns. They discuss assignment language, resource use and software separately. This is evidence of the committee's stated reasons, not causal identification of the decision's effect. In particular, the memorandum's characterization of what the Supreme Court required is not adopted as legal doctrine. The Court itself left the lower court's interpretation of competing assignment language unreviewed (Supreme Court of the United States, 2011, p. 785 n.2).

Hawai'i supplies a contrast between more specific enumeration and continued creator ownership. The 1985 policy reserves qualified copyright ownership to faculty and staff and protects materials prepared on their own initiative for teaching and professional purposes. The 2020 policy retains those central allocations while expressly treating syllabi as own-initiative educational materials. It permits the institution to require copies for specified administrative purposes, but makes publication or reuse depend on consent. This is a more specific allocation of access and use, not a blanket transfer of instructional copyright to the institution. The earlier absence of that explicit reuse condition cannot establish that unauthorized reuse was previously lawful.

The distance-learning provision divides control more visibly. For instruction supported by specified release time or compensation, the university may record or transmit; later sale or retransmission requires a written agreement. The provision must also be read with the work-for-hire language, which expressly includes certain release-time production, and with contractual qualifications. It creates no warrant for treating all recorded teaching as university-owned. Nor is it evidence that distance-learning control first arose with the 2020 wording. Article XI of the 2003–2009 faculty agreement already specifies compensated recording and a written agreement for later sale or rebroadcast (University of Hawaii Professional Assembly and Board of Regents, 2003, pp. 18–19). That agreement additionally specifies a division of net profits and covers faculty rather than every employee. It establishes a prior contractual arrangement, not exact equivalence of the instruments or uninterrupted application to every person.

The Hawai'i endpoints also contain changes outside the newly enumerated teaching categories. The independent-invention exception later expressly requires private personal time, alongside independence from obligations and specified resources. The stated exception is narrower, although the record does not establish how many inventions would change allocation. Conversely, the maximum patent-related publication delay in sponsored-research contracting falls from six months to two months. That changes a stated protection for dissemination rather than initial ownership. These comparisons show why defining the outcome as title alone would omit economically relevant changes in control (University of Hawaii, 1985, IV.2 and IV.8) (University of Hawaii, 2020, III.C.4 and III.C.8).

Table 3 answers the central question at the level the sources support. Utah combines a more

Table 3: Naming knowledge and changing rights: case-level conclusions

Comparison	Definition or enumeration	Allocation conclusion
Utah, 1999–2013	Explicitly enumerated inventions definition adds research outputs and places software under the invention policy; earlier policy already mentions software in commercialization.	Present assignment and revised conditions coexist with a narrower resource branch. Mixed; net expansion indeterminate.
Hawai'i, 1985–2020	Later policy expressly states no unique definitions; additional enumeration of syllabi and distance-learning arrangements is not a formal expansion of the IP definition.	Qualified creator baseline persists; use and exception terms change. Earlier faculty contract limits claims of newly created rights.

Notes: Two selected institutions; no estimated national proportions. Formal definition and category enumeration are recorded separately. Neither comparison is reduced to net institutional expansion. Missing independent validation, contractual scope and intervening changes remain explicit in the underlying register.

explicit definition with changes to assignment and exceptions; Hawai'i adds specific category treatment within a continuing qualified creator baseline and a more complex set of use rights. The latter policy expressly reports no unique definitions, illustrating why category enumeration should not itself be called definitional expansion. Neither case is simply an expansion in vocabulary, but neither supports a single unqualified expansion in institutional ownership. We cannot estimate which pattern is usual across universities from two purposively recovered comparisons.

The broader archive locates these cases without supplying their legal interpretation. Across the 330 adjacent comparisons, research-data terminology appears in 57 previously unmentioned records and disappears in 24; software appears in 48 and disappears in 27 (Table 4). These are endpoint mentions anywhere in the institution-year bundle. An occurrence may identify an exception, refer to commercialization or define a covered object. Utah's earlier software mention is a concrete instance in which a stable lexical indicator would miss a changed place for the category in the operative framework.

Giving each university equal weight leaves positive average changes in research-data and software terminology, but the software result is sensitive to archival restrictions. Restricting gaps to ten years leaves 233 pairs from 97 universities: the software mean is -0.008, and the data mean is 0.050. The replication package retains the full weighting, bootstrap, gap, bundle-size and duplicate-sentence sensitivity outputs. None converts lexical changes into legal changes. A figure purporting to show historical ownership shares would therefore be unsupported, as would a population table that treated unreviewed arrangements as absent.

The other recovered policies reinforce the relational interpretation without adding authenticated historical transitions. Creighton's 2019 policy combines a resource-conditioned claim, reciprocal

Table 4: Within-university transitions in category mentions

Category	0 to 0	0 to 1	1 to 0	1 to 1
Algorithms	311	8	3	8
Biological materials	179	45	21	85
AI/mask-work/semiconductor terms	246	29	16	39
Inventions/patents	1	3	3	323
Research data/databases	151	57	24	98
Scholarly/publication terms	13	20	11	286
Software	56	48	27	199
Tangible research property	224	37	12	57
Teaching materials	126	54	24	126

Notes: Each row describes the same 330 adjacent observed-vintage pairs. Zero and one denote absence and presence of a lexical flag anywhere in the institution-year bundle. Categories overlap. These are not distributions of ownership arrangements. The final row combines several technologies and does not date their emergence.

nonexclusive use, academic exclusions and an assignment form (Creighton University, 2019, sections C, D.2 and Appendix I). Illinois State's 2024 policy protects some educational software while separately addressing employee inventions and student coursework (Illinois State University, 2024, sections 2.9, 2.12 and 3.1–3.2). Wake Forest's earlier retrieved policy is a 2006-amended reprint, and its 2017 corpus bundle combines campus-specific instruments; these remain outside the substantive paired results (Wake Forest University, 2006, 2017). Palmer's account of Caltech corroborates a historical conditional-allocation structure, but its modal wording differs from the cleaned 1945 record (Palmer, 1962, pp. 55–57). Recovery strengthens the evidence only where version and scope can be established for the particular comparison.

6 Validation, competing explanations and limits

Automated text analysis requires validation for the quantity being measured (Grimmer and Stewart, 2013; Gentzkow et al., 2019). Agreement between two keyword systems would not validate a legal interpretation, and agreement among repeated AI outputs would not constitute independent human annotation. We provide a stratified sample of 178 sentences, including both screen-positive and screen-negative cases. It covers decades and 68 institutions, with known inclusion probabilities. Category coverage is reported separately because rare categories are not guaranteed adequate representation. The annotation template remains unfilled pending a credible contextual reference standard.

The selected source readings and paired relationship interpretations are a different exercise. They were chosen to investigate apparent assignment, exception, license and historical-context problems, and their interpretations are explicitly AI-assisted. They show actual failures of the original

retrieval logic: a defined term can replace an asset name, operative conditions can occur in the next sentence, and a retained license can matter without transferring title. They do not supply a probability sample of errors. We therefore report no precision, recall, kappa or category accuracy. Agreement statistics require appropriate annotation data and assumptions; they are not a substitute for correctness (Artstein and Poesio, 2008).

A complete validation would link each sampled provision to an authenticated version, have independent qualified readers identify its relationships and qualifications, retain disagreements, and adjudicate them transparently. Negative screen cases are essential to recall estimation. Rare categories require additional sampling with recorded probabilities, rather than an unweighted collection of convenient examples. Context-unverified provisions should remain unknown, including when a sentence appears linguistically unambiguous. The released codebook specifies these rules so that later validation can change labels without changing the frozen population.

Several explanations compete with ownership expansion. New terminology may clarify an old umbrella claim; a consolidated policy may add previously separate documents; a new exception may introduce the very category it protects. Longer policies create more opportunities for a lexical match. Changes in digitization or sentence cleaning may alter what survives in the corpus. System-wide policies repeated across campuses can give one legal instrument several apparent institutional observations. These mechanisms are consistent with some of the observed lexical changes and cannot be distinguished by their direction alone.

Within-university comparisons address one concern but leave others intact. They avoid interpreting a later sample of different universities as institutional change, yet still require comparable documents at both endpoints. A university may separate its medical and general-campus rules, replace a policy with guidance, or publish a form alongside a substantive instrument. The evidence required to call a transition is therefore more demanding than institution identity plus chronological ordering. A common actor and fact pattern must face a different rule under two authenticated versions. Where that comparison cannot be made, an unresolved transition is the accurate observation.

The source recovery undertaken here is necessarily partial. Accessible institutional sources include selected historical policy endpoints, a committee memorandum, a faculty agreement, contemporary provisions, an earlier reprint and a historical survey. They are not a recovered archive of all underlying policies. The cleaned corpus lacks a complete source-URL manifest, some links fail, and official current pages cannot establish their earlier contents merely because they are authoritative today. We preserve successful retrieval hashes and failed attempts. Reconstructed sentence context is useful for review but is never presented as original paragraph context.

Finally, neither this archive nor the selected policies establish enforceability. We do not observe all signed agreements, sponsored-research contracts, state-law restrictions or disputes. A university may announce a claim that would fail in a particular case, or possess a contractual right not fully

expressed in a public policy. The article concerns stated institutional allocation and the evidence required to measure its evolution. It does not equate those statements with the legal rights ultimately recognized by a court.

7 Discussion and conclusion

The source-linked comparisons change the interpretation of policy expansion. A university can strengthen the language of assignment while narrowing a resource-based claim. It can describe new instructional uses while preserving creator ownership and conditioning subsequent institutional use on agreement. An apparently new policy arrangement may also have a predecessor in a collective agreement. The observations concern different instruments of control, not merely a longer list of intellectual property categories.

For institutional economics, the implication is a measurement requirement with substantive consequences. Title, authority to commercialize, permission to use and constraints on publication allocate different decisions. A transfer-mechanism change can matter without a change in the nominal owner; a publication-delay ceiling can change while ownership stays fixed. The Utah and Hawai'i comparisons demonstrate such distinctions within institutions. They do not establish the welfare consequences of those choices, whether the stated rules changed behavior, or which competing interests prevailed in particular negotiations.

The evidence also limits a technological explanation of expanding policy language. New forms of work can require more specific rules, but changes in legal drafting and the movement of arrangements across institutional documents can produce similar textual patterns. Utah's committee provides direct evidence of several stated motivations. Hawai'i's earlier faculty agreement demonstrates that an executive-policy date need not date the underlying arrangement. These findings support attention to institutional rulemaking and contractual context; they do not identify imitation, technological change or judicial decisions as general causal mechanisms.

The national historical question remains open. The fixed archive supports reproducible within-university comparisons of terminology, while the recovered sources support a small set of substantive relationship comparisons. They cannot establish whether most universities broadened claims when they broadened definitions. A defensible framing is therefore a comparative historical study of how definitions, assignment and creator protections interact, with a larger archive documenting the measurement and selection problem. A population history requires more comparable sources and independent contextual coding. The present positive conclusion is narrower but consequential: changes in university control can be multidirectional within a single institution, and neither new vocabulary nor nominal ownership alone captures them.

Data, code and research transparency

The repository contains immutable source identifiers and checksums, construction and analysis scripts, deterministic sampling, derived results, a legal codebook, an annotation template and a source-linked evidence ledger. Raw text and downloaded policy documents are retrieved locally rather than newly redistributed in this package; third-party rights have not been comprehensively assessed. Each table and figure is linked to its generating output. The canonical manuscript is synchronized into a self-contained Overleaf export. Author identity, affiliations, funding and conflict declarations must be supplied and confirmed before submission.

AI assistance was used in programming, source discovery, bibliographic comparison, close reading and drafting. The interpretations attributed to that process have not been independently validated by human legal coders. The authors must review and assume responsibility for the manuscript before submission. No human coding, legal review or endorsement is represented as completed by this statement.

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